

# Geometry-based Methods in Vision

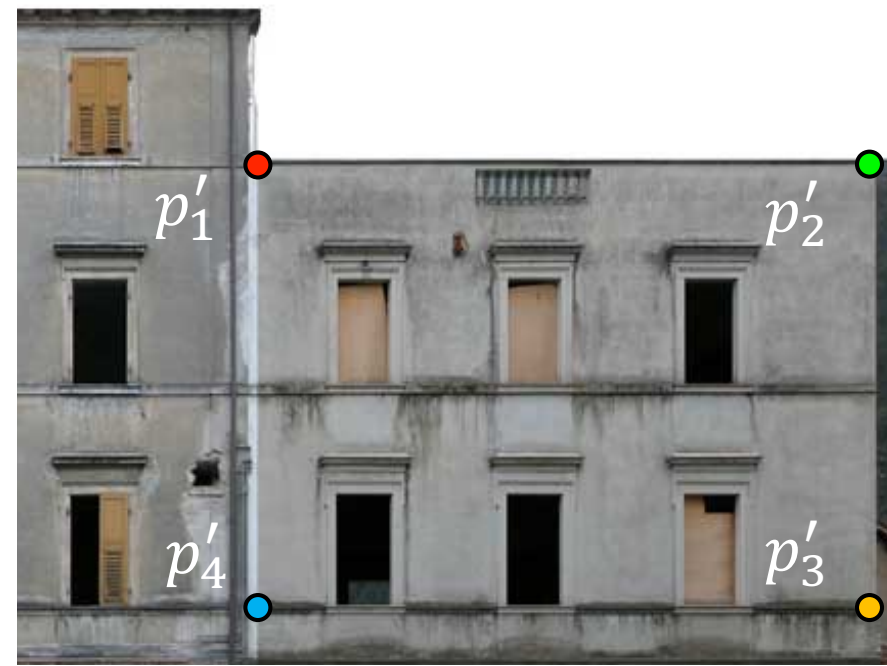
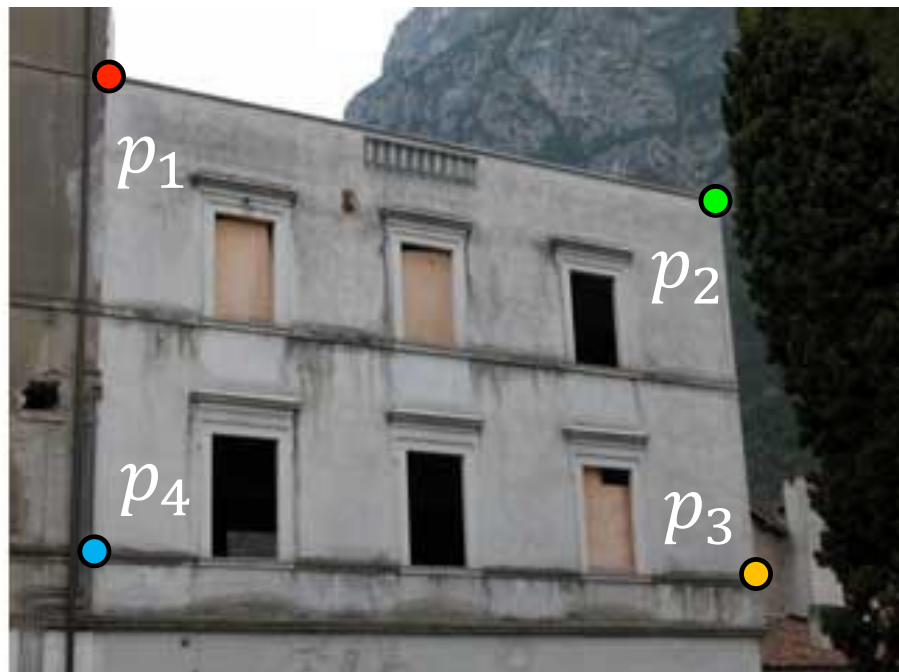
CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

# Administrivia

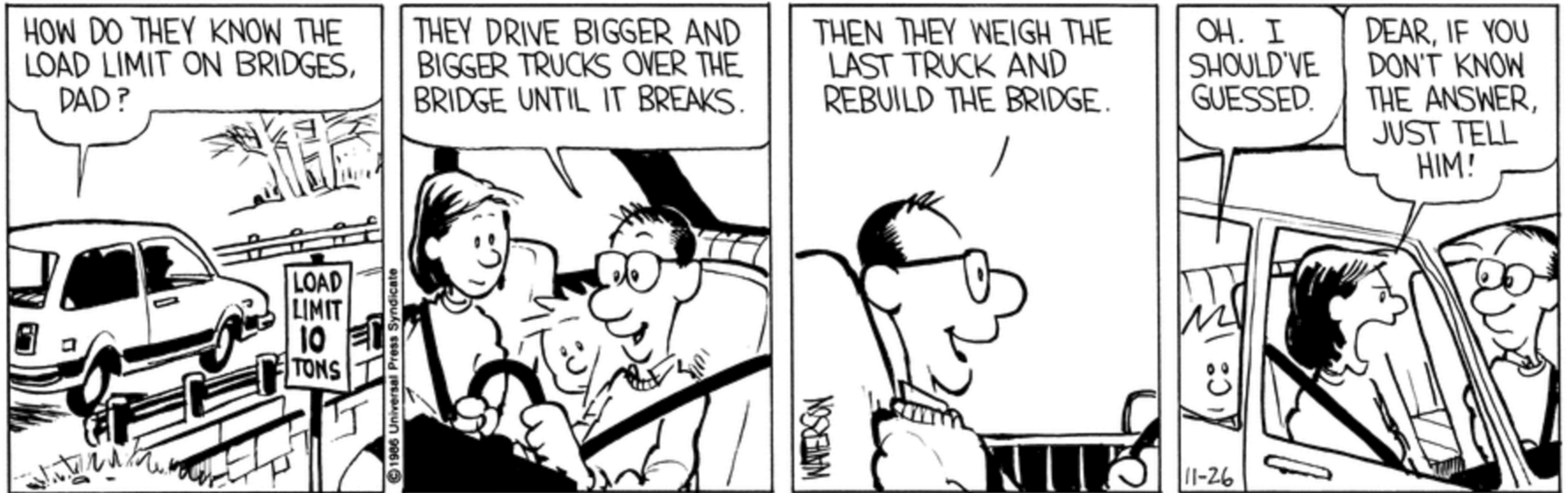
- PS1 out today
  - *Due on Tue in 1 week*
- Assignment 1 out today
  - *Due on Tue in 2 weeks*
- PS Submission Instructions
  - *Please use the provided template to hand-write answers*
  - *We will modify it to give you more space*

# Lecture 5: Homography from Correspondences





# A reminder to ask questions!

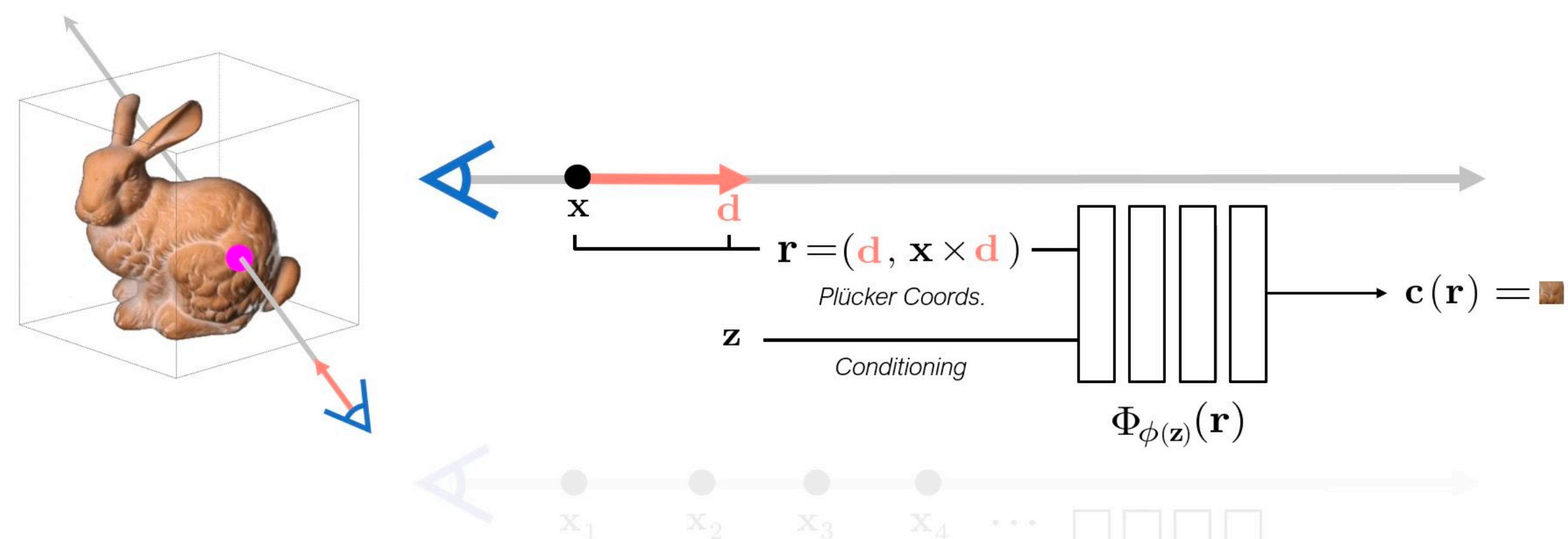


# Lines in 3D Projective Geometry

$$L \equiv \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T \equiv \begin{bmatrix} 0 & l_{12} & l_{13} & l_{14} \\ -l_{12} & 0 & l_{23} & l_{24} \\ -l_{13} & -l_{23} & 0 & l_{34} \\ -l_{14} & -l_{24} & -l_{34} & 0 \end{bmatrix}$$

Plucker representation

$$\mathbf{x}_3 = \mathbf{x}_1 + \beta \mathbf{x}_2 \quad \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T = \mathbf{x}_1 \mathbf{x}_3^T - \mathbf{x}_3 \mathbf{x}_1^T \quad \text{Invariant to the points used to compute!}$$



Exercise: Show that the representation  $(\tilde{\mathbf{d}}, \tilde{\mathbf{x}} \times \tilde{\mathbf{d}})$  is equivalent to the plucker representation above

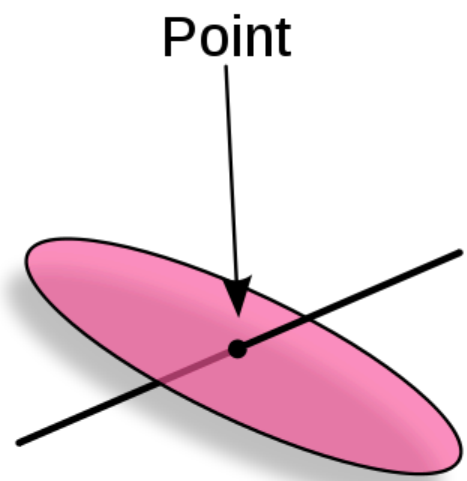
Useful as input representation in modern NNs!



# Lines in 3D Projective Geometry

$$L \equiv \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T \equiv \begin{bmatrix} 0 & l_{12} & l_{13} & l_{14} \\ -l_{12} & 0 & l_{23} & l_{24} \\ -l_{13} & -l_{23} & 0 & l_{34} \\ -l_{14} & -l_{24} & -l_{34} & 0 \end{bmatrix}$$

Plucker representation



Point given a line and a plane:

$$\mathbf{x} = L\pi$$

Q: what constraint if a line is parallel to a plane?

Q: what constraint if a line lies in a plane?

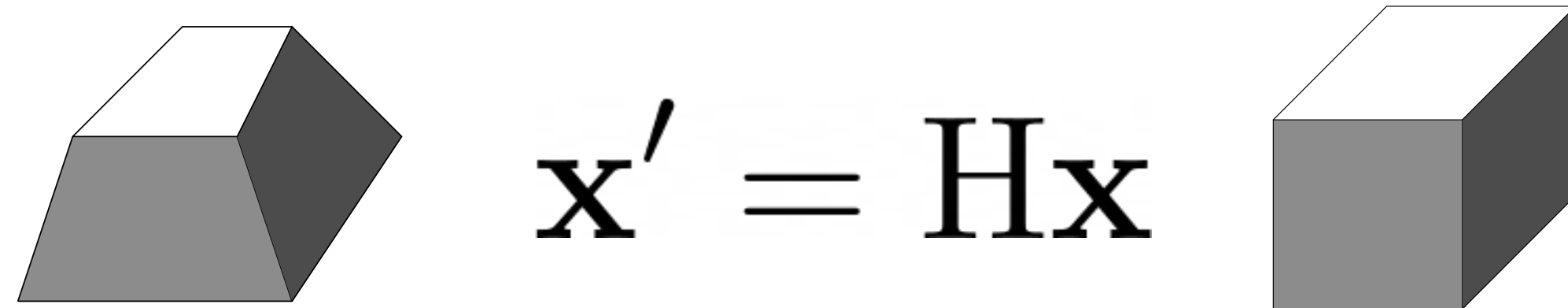
Plane containing a line and point:

$$\pi = L^* \mathbf{x}$$

Q: what constraint if a point lies on the line?

$$L^* \equiv \begin{bmatrix} 0 & l_{34} & l_{24} & l_{23} \\ -l_{34} & 0 & l_{14} & l_{13} \\ -l_{24} & -l_{14} & 0 & l_{12} \\ -l_{23} & -l_{13} & -l_{12} & 0 \end{bmatrix}$$

# 3D Projective Geometry: Transformations



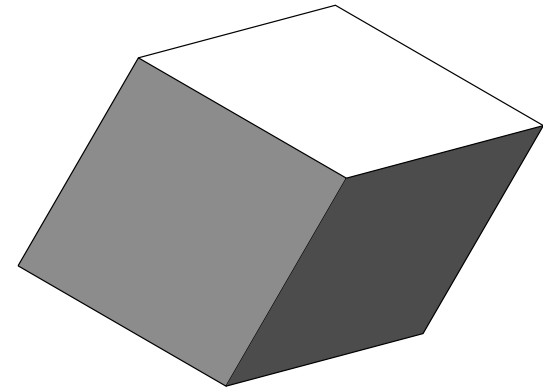
Projectivity/Homography: **H** is an invertible matrix

Q: How many degrees of freedom for **H** ?

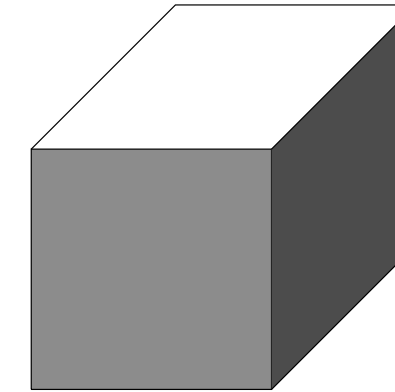
Q: How does a plane transform under **H** ?

$$\pi' = \mathbf{H}^{-\top} \pi$$

# Hierarchy of 3D Transforms: Euclidean



$$\mathbf{x}' = \mathbf{H}_e \mathbf{x} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{x}$$

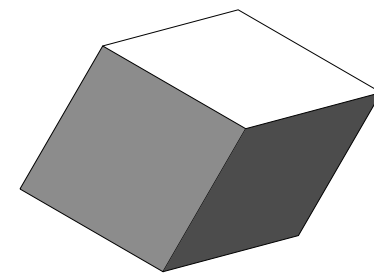


All distances are preserved

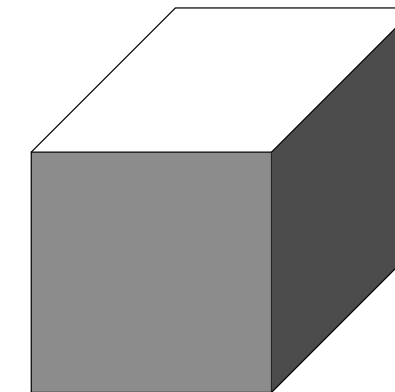
Q: How many degrees of freedom?



# Hierarchy of 3D Transforms: Similarity



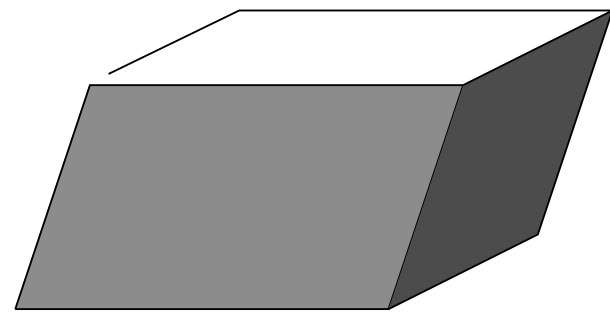
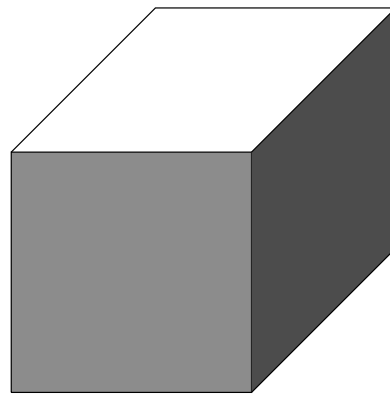
$$\mathbf{x}' = \mathbf{H}_s \mathbf{x} = \begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{x}$$



Angles are preserved

7 DoF transform

# Hierarchy of 3D Transforms: Affine


$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{x}$$


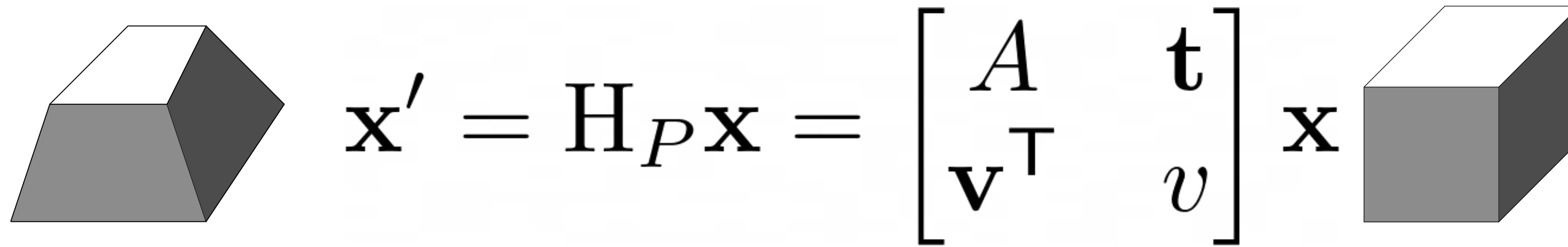
Parallelism is preserved

Q: Degrees of freedom?

Q: What structure would remain invariant?

$$\pi_{\infty} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

# Hierarchy of 3D Transforms: (General) Projective

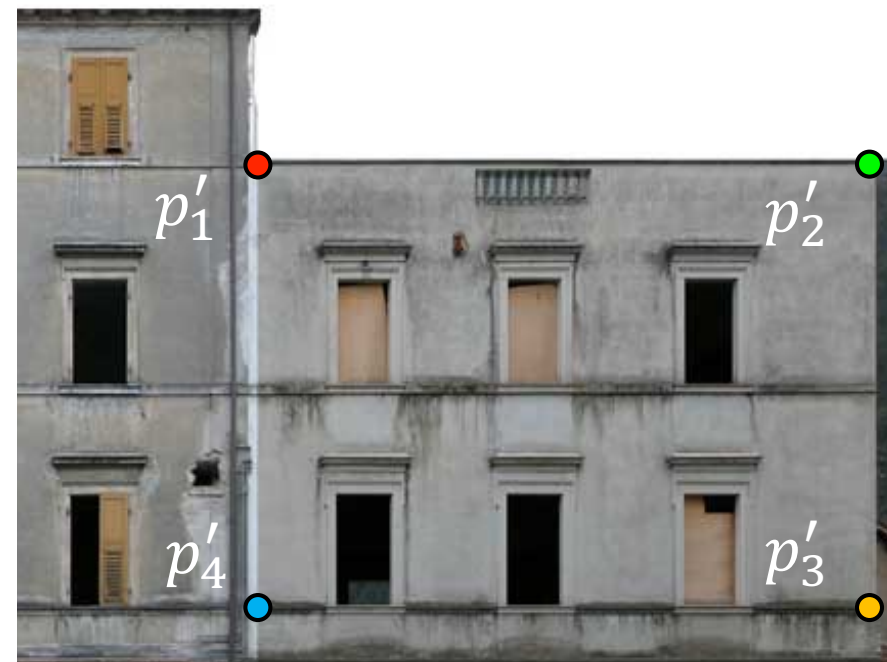
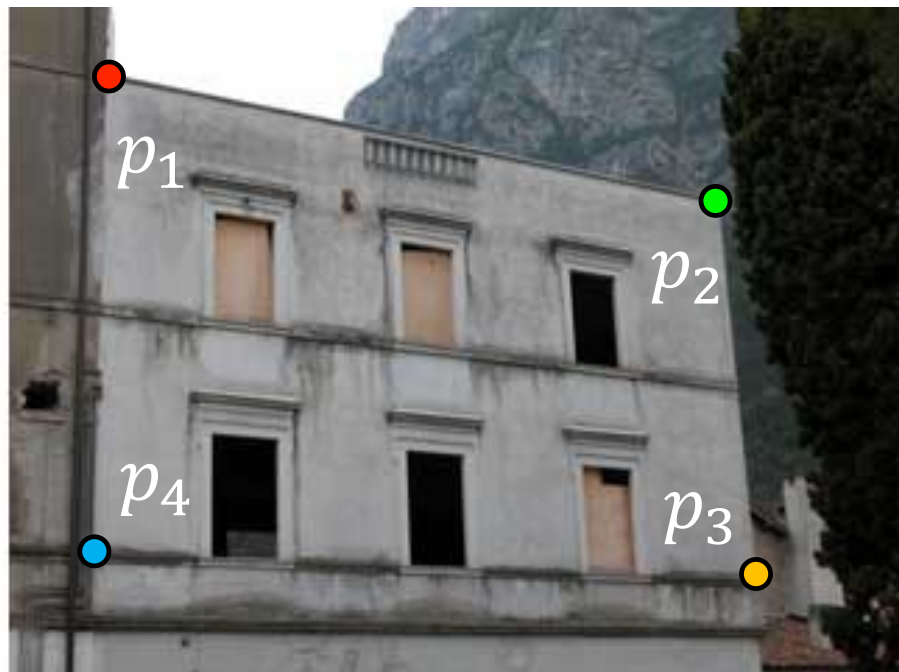

$$\mathbf{x}' = \mathbf{H}_P \mathbf{x} = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix} \mathbf{x}$$

Q: Degrees of freedom?

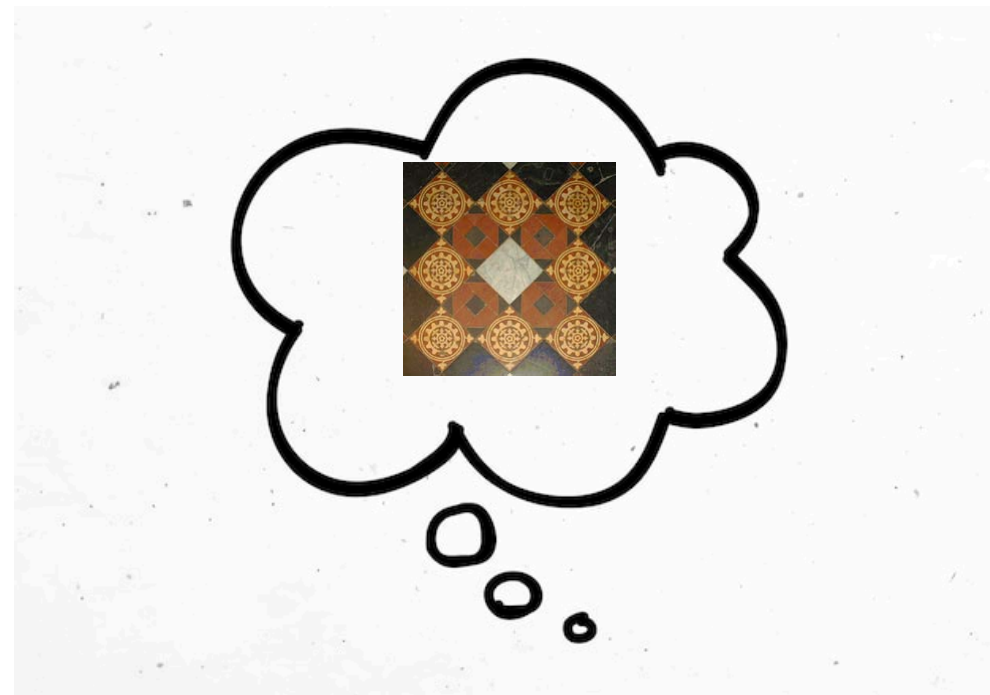
Decompositions similar to the 2D case apply  
(will revisit when studying camera models)



# Lecture 5: Homography from Correspondences



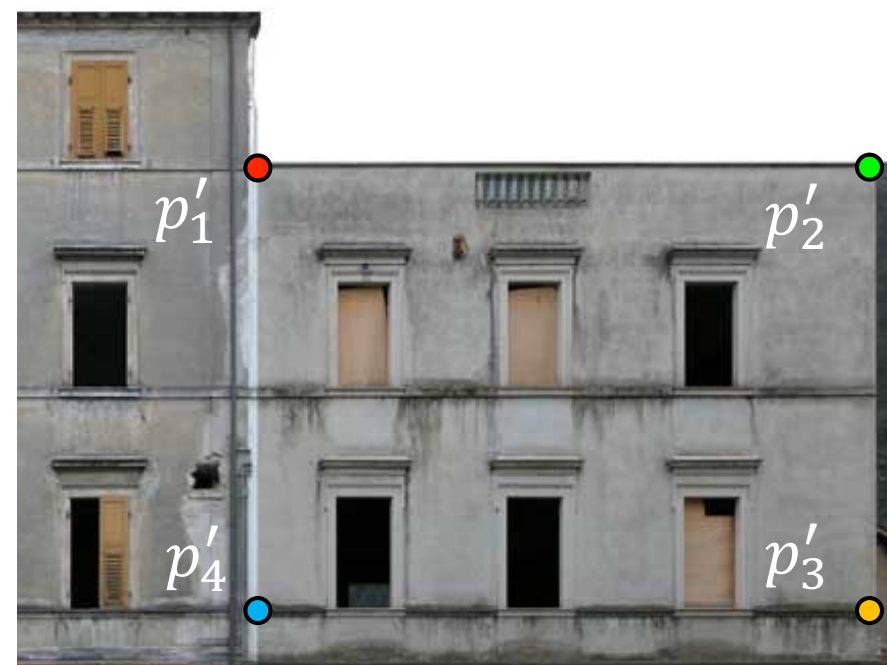
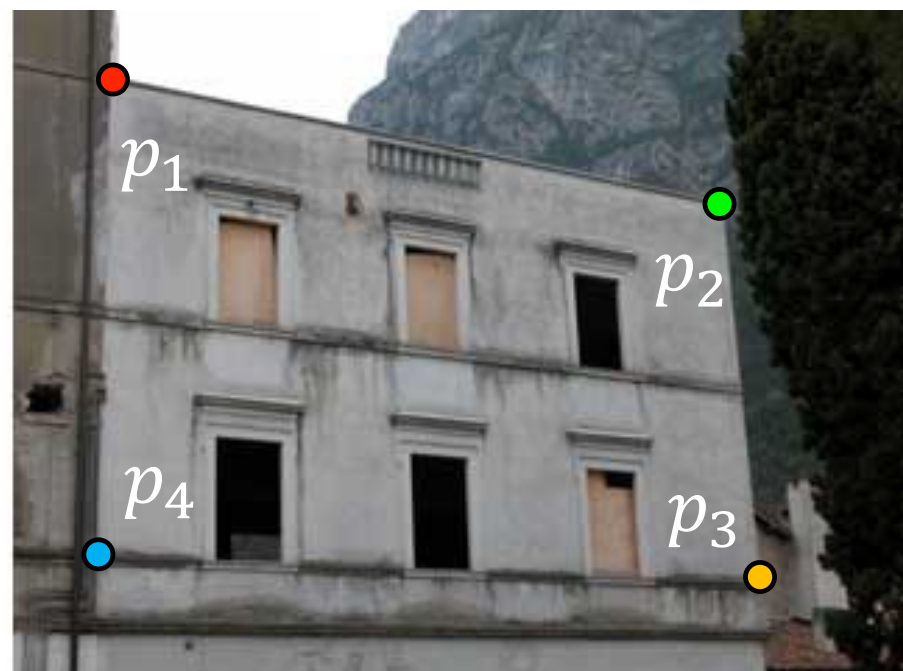
# Estimating Projective Transforms



$H$

Computing **H** from a **single** image.

Used annotations about metric or affine properties.



$H$

Computing **H** to align a **pair** of images

Use annotated/automatic correspondences.

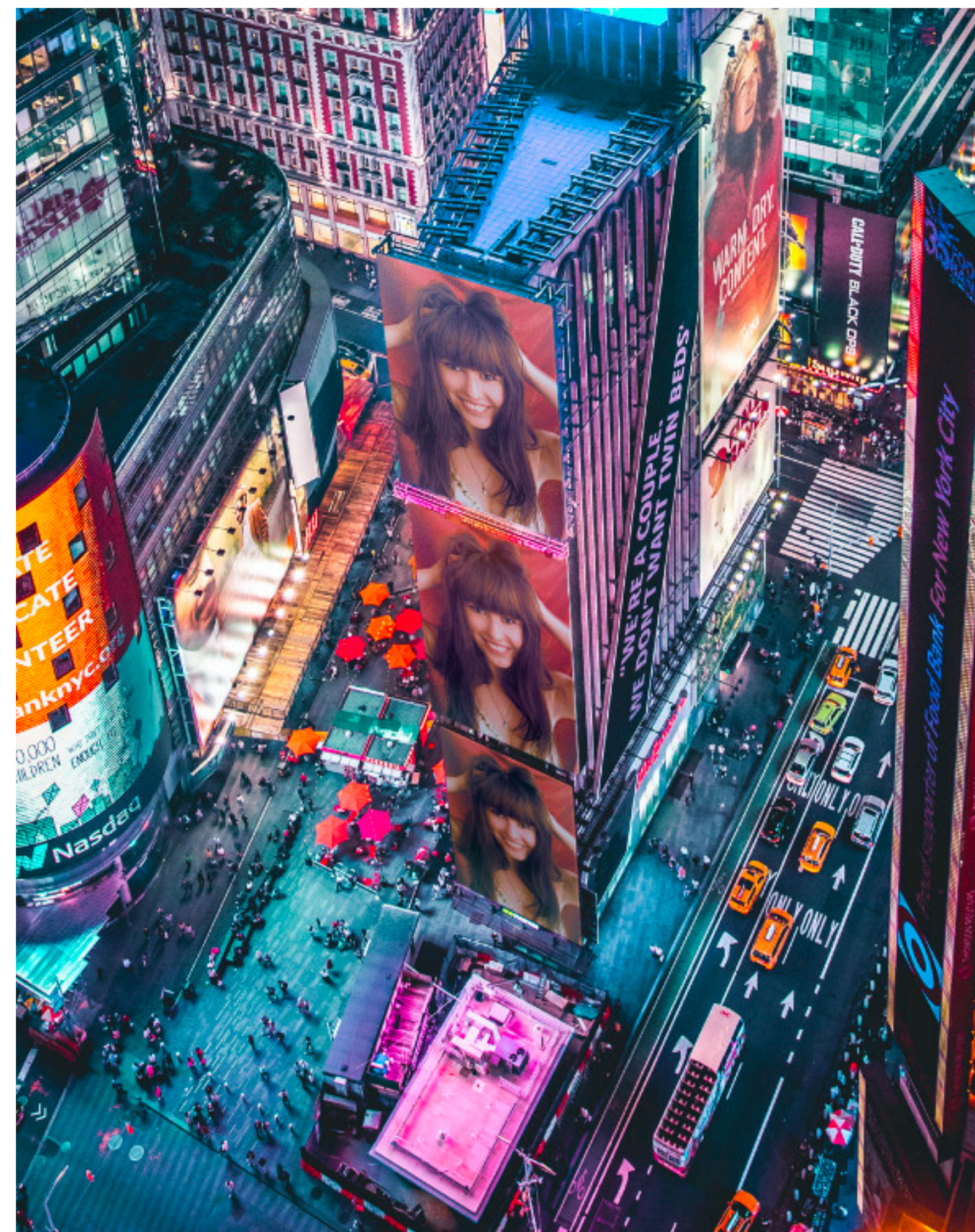
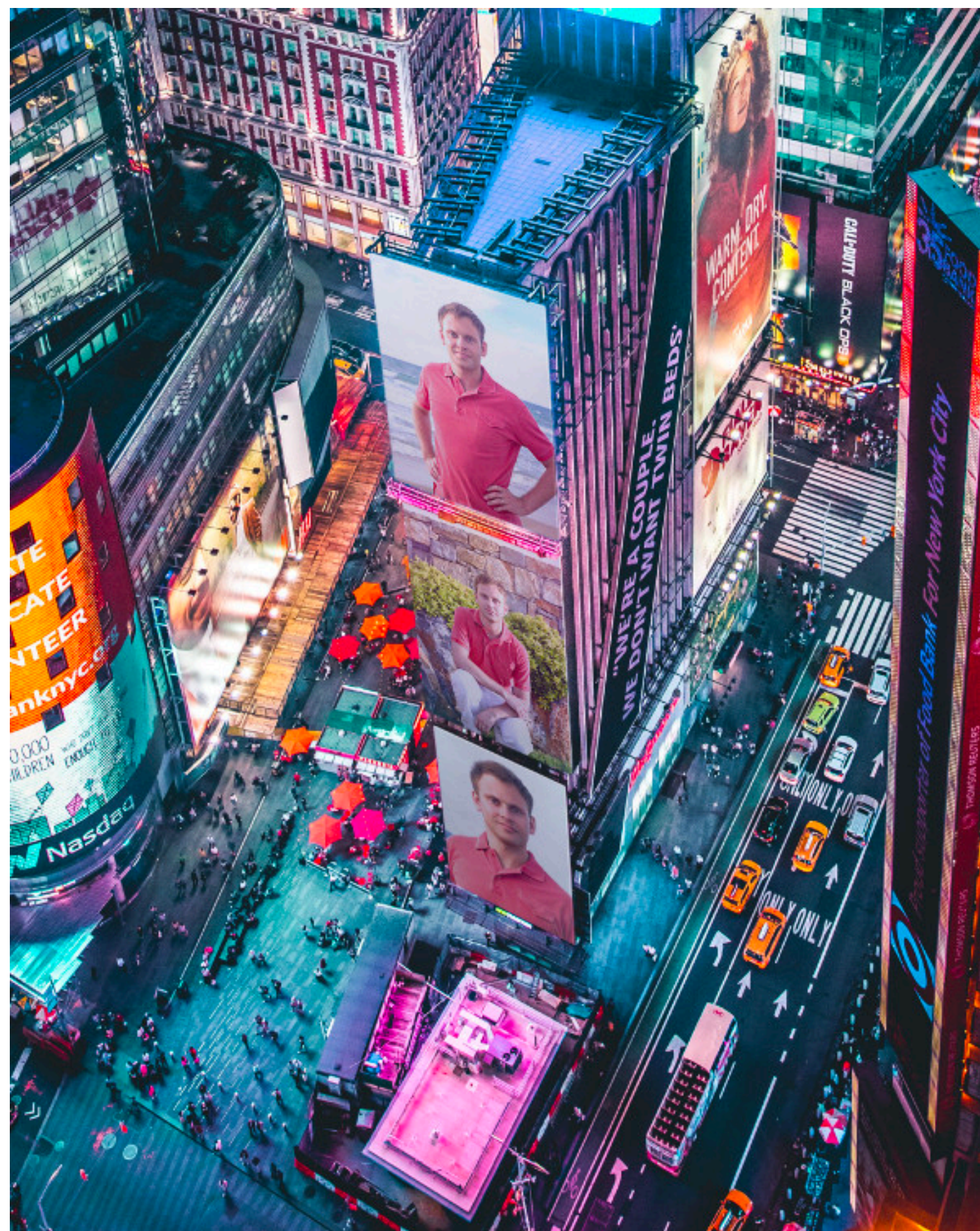


# Projective Transforms between Images



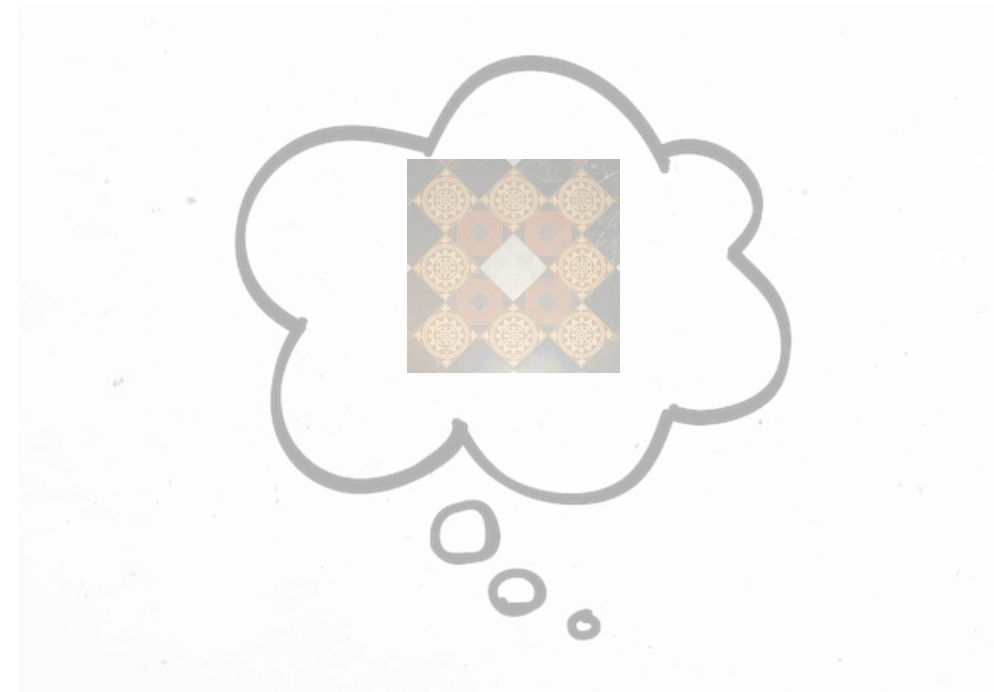


# Projective Transforms between Images





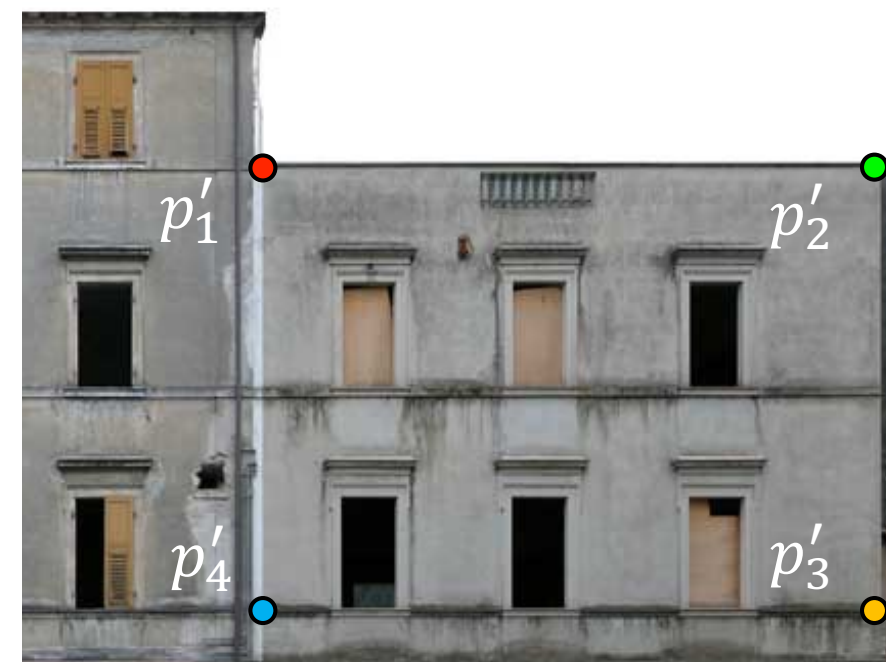
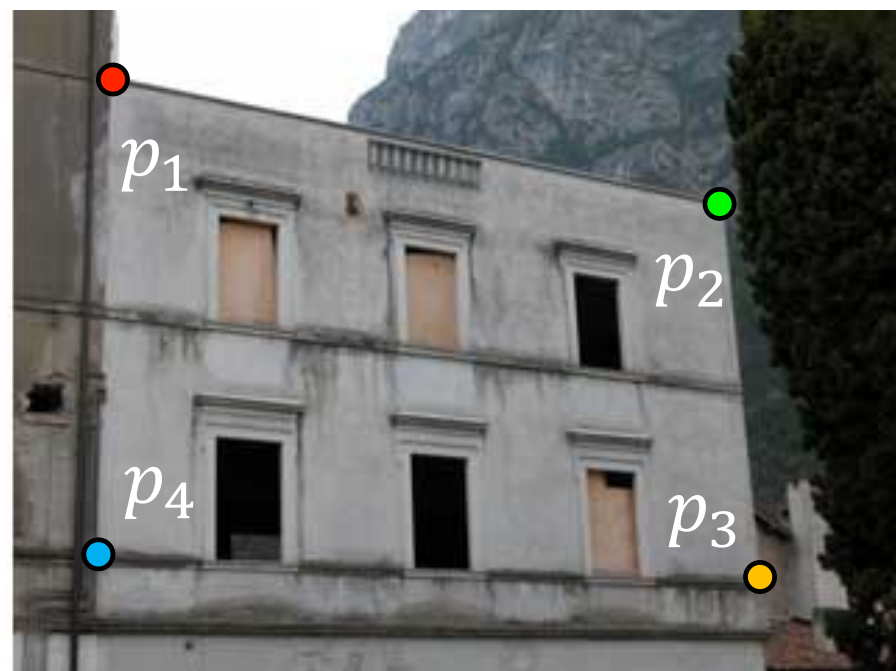
# Estimating Projective Transforms



$H$

Computing **H** from a **single** image.

Used annotations about metric or affine properties.

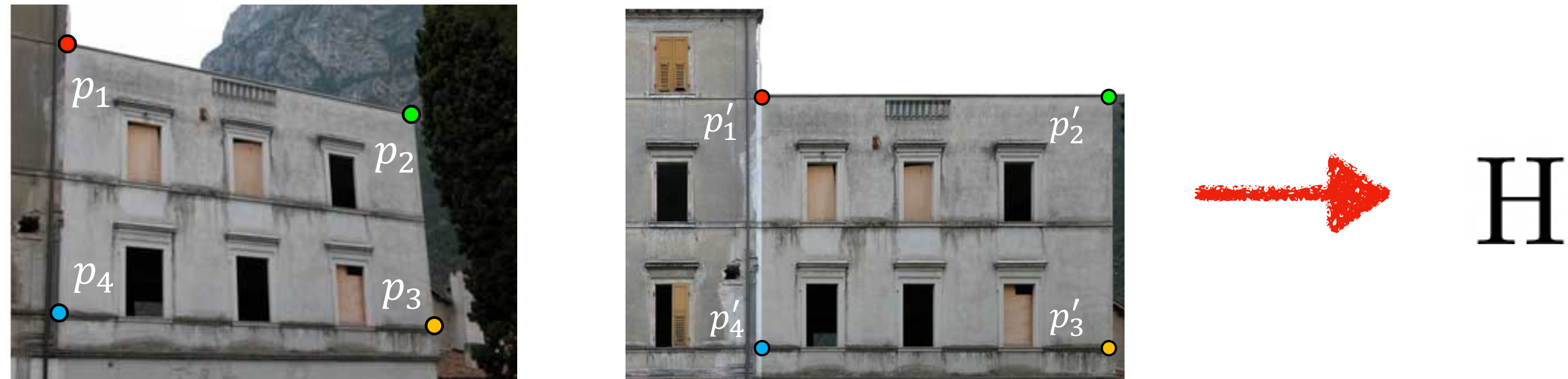


$H$

Computing **H** to align a **pair** of images

Use annotated/automatic correspondences.

# Estimating Homographies from Correspondences



Goal: Given pairs  $(\mathbf{x}_i, \mathbf{x}'_i)$ , find  $\mathbf{H}$  s.t.  $\mathbf{H}\mathbf{x}_i = \mathbf{x}'_i$

Approach:

~~$$x_i^1 \mathbf{h}_1 + x_i^2 \mathbf{h}_2 + x_i^3 \mathbf{h}_3 = \mathbf{x}'_i$$~~

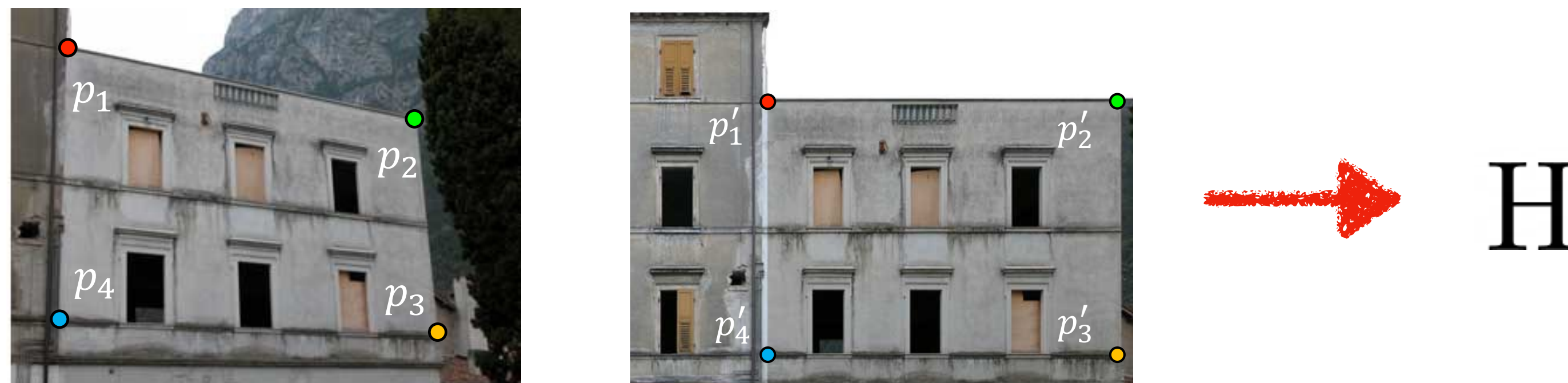
~~Each correspondence gives  
3 linear constraints~~

why?

~~$$A\mathbf{h} = \mathbf{b}$$~~

~~Using  $\geq 3$  constraints, can  
solve for  $\mathbf{H}$~~

# Estimating Homographies from Correspondences



Goal: Given pairs  $(\mathbf{x}_i, \mathbf{x}_i')$ , find  $\mathbf{H}$  s.t.  $\mathbf{H}\mathbf{x}_i = \mathbf{x}_i'$

Q: what would imply that  $\mathbf{H}\mathbf{x}_i$  and  $\mathbf{x}_i'$  are (projectively) equal?

$$H\mathbf{x}_i \times \mathbf{x}'_i = \mathbf{0}$$

Q: If  $\mathbf{x}=\mathbf{y}$ , what constraint should hold in higher dimensions ?

$$\mathbf{y}\mathbf{x}^\top - \mathbf{x}\mathbf{y}^\top = 0$$

A 'generalization' of cross product.

Where did we see this?



# Estimating Homographies from Correspondences

$$H\mathbf{x} \times \mathbf{x}' = \mathbf{0}$$

$$H = \begin{bmatrix} \mathbf{h}^{1\top} \\ \mathbf{h}^{2\top} \\ \mathbf{h}^{3\top} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{h}^1 \\ \mathbf{h}^2 \\ \mathbf{h}^3 \end{bmatrix} \quad H\mathbf{x} = \begin{bmatrix} \mathbf{x}^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{x}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{x}^\top \end{bmatrix} \mathbf{h}$$

From Lecture 2:

$$\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$
$$\mathbf{a} \times \mathbf{b} = [\mathbf{b}]_{\times}^{\top} \mathbf{a} = \begin{bmatrix} 0 & b_3 & -b_2 \\ -b_3 & 0 & b_1 \\ b_2 & -b_1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix},$$

Cool trick (really useful later): cross product as matrix multiplication!

# Estimating Homographies from Correspondences

$$H\mathbf{x} \times \mathbf{x}' = \mathbf{0}$$

$$H = \begin{bmatrix} \mathbf{h}^{1\top} \\ \mathbf{h}^{2\top} \\ \mathbf{h}^{3\top} \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} \mathbf{h}^1 \\ \mathbf{h}^2 \\ \mathbf{h}^3 \end{bmatrix} \quad H\mathbf{x} = \begin{bmatrix} \mathbf{x}^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{x}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{x}^\top \end{bmatrix} \mathbf{h}$$

$$[\mathbf{x}']^\top \times \begin{bmatrix} \mathbf{x}^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{x}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{x}^\top \end{bmatrix} \mathbf{h} = \mathbf{0}$$

Linear constraint over the elements of  $\mathbf{h}$

# Estimating Homographies from Correspondences

$$[\mathbf{x}']_{\times}^{\top} \begin{bmatrix} \mathbf{x}^{\top} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{x}^{\top} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{x}^{\top} \end{bmatrix} \mathbf{h} = \mathbf{0}$$

Q: How many constraints does each correspondence yield?

$$\begin{bmatrix} \mathbf{0} & -w'\mathbf{x}^{\top} & y'\mathbf{x}^{\top} \\ w'\mathbf{x}^{\top} & \mathbf{0} & -x'\mathbf{x}^{\top} \\ -y'\mathbf{x}^{\top} & x'\mathbf{x}^{\top} & \mathbf{0} \end{bmatrix} \mathbf{h} = \mathbf{0}$$

Exercise: show that the third constraint is linearly dependent on first two



# Estimating Homographies from Correspondences

$$H\mathbf{x} \times \mathbf{x}' = \mathbf{0}$$

$$H \equiv \begin{bmatrix} h^{1\top} \\ h^{2\top} \\ h^{3\top} \end{bmatrix} \quad \mathbf{h} \equiv \begin{bmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix} \quad \begin{bmatrix} \mathbf{0} & -w'\mathbf{x}^\top & y'\mathbf{x}^\top \\ w'\mathbf{x}^\top & \mathbf{0} & -x'\mathbf{x}^\top \\ -y'\mathbf{x}^\top & x'\mathbf{x}^\top & \mathbf{0} \end{bmatrix} \mathbf{h} = \mathbf{0}$$

Each correspondence yields two constraints of the form  $\mathbf{A}_i \mathbf{h} = \mathbf{0}$

Given  $\geq 4$  correspondences, solve for  $\mathbf{h}$ :  $\min_{\mathbf{h}} \|\mathbf{A}\mathbf{h}\|$

Q: is some constraint missing?  $\|\mathbf{h}\| = 1$

# Estimating Homographies from Correspondences

Q: What if H is known to be affine?

$$H\mathbf{x} \times \mathbf{x}' = \mathbf{0}$$

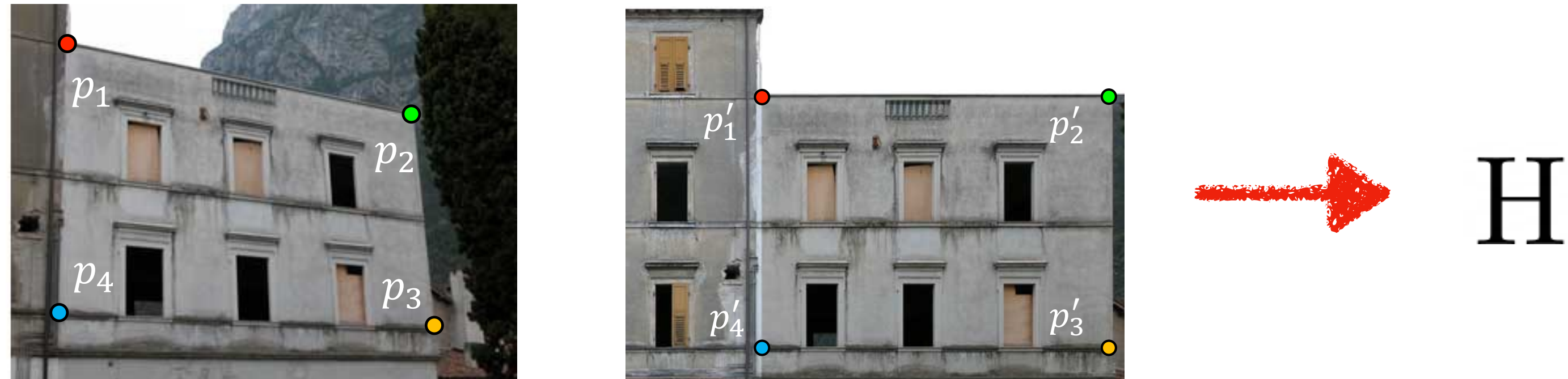
$$H \equiv \begin{bmatrix} h^{1\top} \\ h^{2\top} \\ h^{3\top} \end{bmatrix} \quad \mathbf{h} \equiv \begin{bmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix} \quad \begin{bmatrix} \mathbf{0} & -w'\mathbf{x}^\top & y'\mathbf{x}^\top \\ w'\mathbf{x}^\top & \mathbf{0} & -x'\mathbf{x}^\top \\ -y'\mathbf{x}^\top & x'\mathbf{x}^\top & \mathbf{0} \end{bmatrix} \mathbf{h} = \mathbf{0}$$

$\mathbf{h}_3 = [0 \ 0 \ 1]^\top$

Each correspondence yields two constraints of the form  $\mathbf{A}_i \mathbf{h} = \mathbf{0}$

Given  $\geq 3$  correspondences, solve for  $\mathbf{h}$ :  $\min_{\mathbf{h}} \|\mathbf{A}\mathbf{h}\|$   
 $\|\mathbf{h}\| = 1$

# Estimating Homographies from Correspondences



Do *any* 4 points suffice?

Q: Given correspondences for  $\mathbf{p}_1$  and  $\mathbf{p}_2$ , what other points do not provide 2 additional constraints for  $\mathbf{H}$ ?

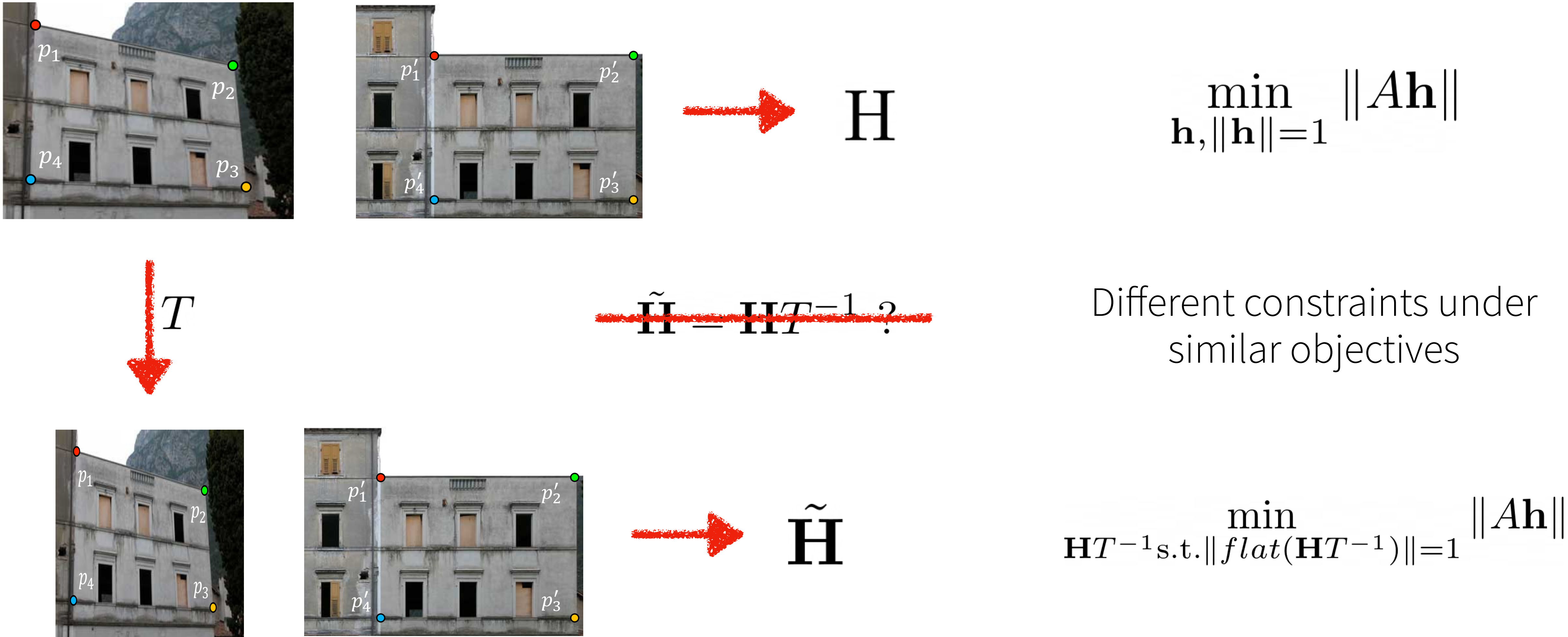
No 3 points should be collinear!

If  $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$  lie on  $\mathbf{l}$  —

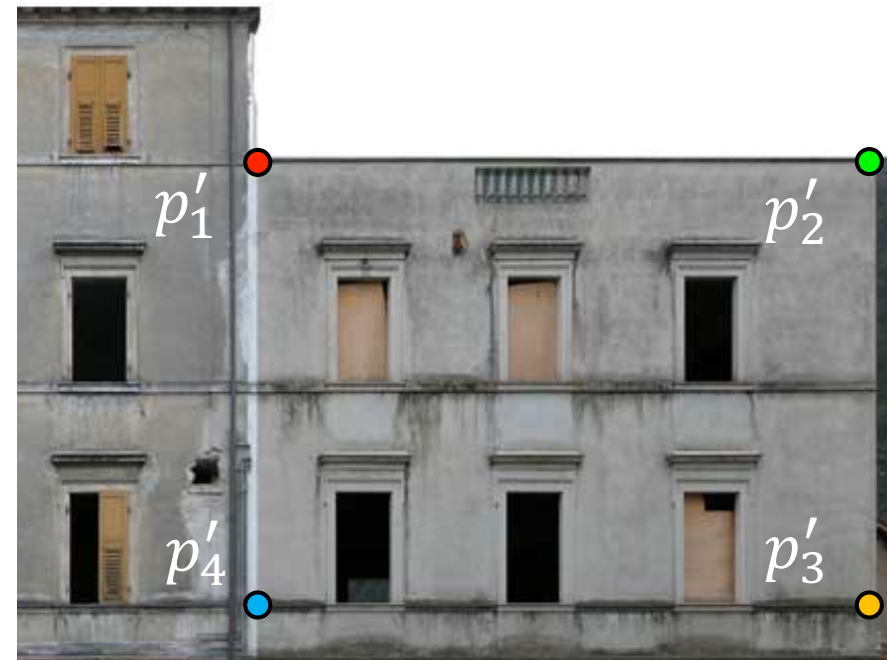
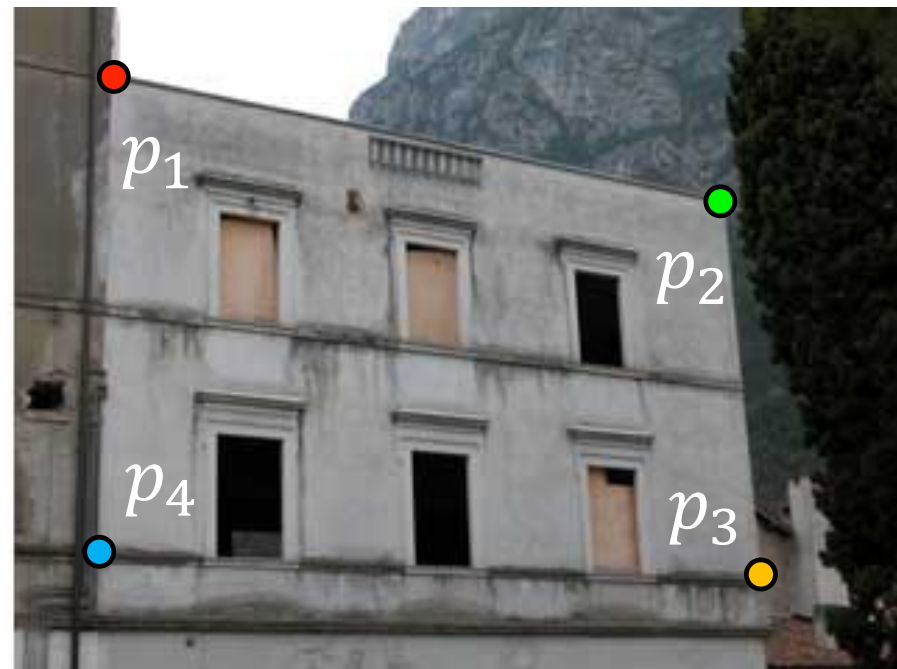
$$H^* = \mathbf{x}'_4 \mathbf{l}^\top$$



# Non-invariance of DLT



# Practical DLT: Normalization



$\mathbf{H}$

$$\min_{\mathbf{h}, \|\mathbf{h}\|=1} \|\mathbf{A}\mathbf{h}\|$$

$$\begin{bmatrix} \mathbf{0} & -w'\mathbf{x}^\top & y'\mathbf{x}^\top \\ w'\mathbf{x}^\top & \mathbf{0} & -x'\mathbf{x}^\top \\ -y'\mathbf{x}^\top & x'\mathbf{x}^\top & \mathbf{0} \end{bmatrix} \mathbf{h} = \mathbf{0}$$

SVD yields a low-rank approximation minimizing euclidean error

**A** includes terms of the form  $\mathbf{x}'\mathbf{x}$ ,  $\mathbf{w}'\mathbf{w}$ , ....

very different possible ranges given scale of pixel coordinates

# Practical DLT: Putting it Together

## Objective

Given  $n \geq 4$  2D to 2D point correspondences  $\{\mathbf{x}_i \leftrightarrow \mathbf{x}'_i\}$ , determine the 2D homography matrix  $\mathbf{H}$  such that  $\mathbf{x}'_i = \mathbf{H}\mathbf{x}_i$ .

## Algorithm

- (i) **Normalization of  $\mathbf{x}$ :** Compute a similarity transformation  $\mathbf{T}$ , consisting of a translation and scaling, that takes points  $\mathbf{x}_i$  to a new set of points  $\tilde{\mathbf{x}}_i$  such that the centroid of the points  $\tilde{\mathbf{x}}_i$  is the coordinate origin  $(0, 0)^\top$ , and their average distance from the origin is  $\sqrt{2}$ .
- (ii) **Normalization of  $\mathbf{x}'$ :** Compute a similar transformation  $\mathbf{T}'$  for the points in the second image, transforming points  $\mathbf{x}'_i$  to  $\tilde{\mathbf{x}}'_i$ .
- (iii) **DLT:** Apply algorithm 4.1(p91) to the correspondences  $\tilde{\mathbf{x}}_i \leftrightarrow \tilde{\mathbf{x}}'_i$  to obtain a homography  $\tilde{\mathbf{H}}$ .
- (iv) **Denormalization:** Set  $\mathbf{H} = \mathbf{T}'^{-1}\tilde{\mathbf{H}}\mathbf{T}$ .



# Beyond DLT

$$L = \sum_i d(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i) \quad \min_H L$$

$$d_{\text{alg}}(\mathbf{x}, \mathbf{x}') = \|\mathbf{x}' \times \mathbf{x}\|^2$$

Need additional constraint to avoid degenerate solution

Closed form solution

Non-equivariant under similarity transform

$$d_{\text{geom}}(\mathbf{x}, \mathbf{x}') = \|x'/w' - x/w\|^2 + \|y'/w' - y/w\|^2$$

No additional constraint needed

(Pixel projection loss)

Not linear in  $\mathbf{h}$

Equivariant to similarity transform

Approach to using non-linear objectives:

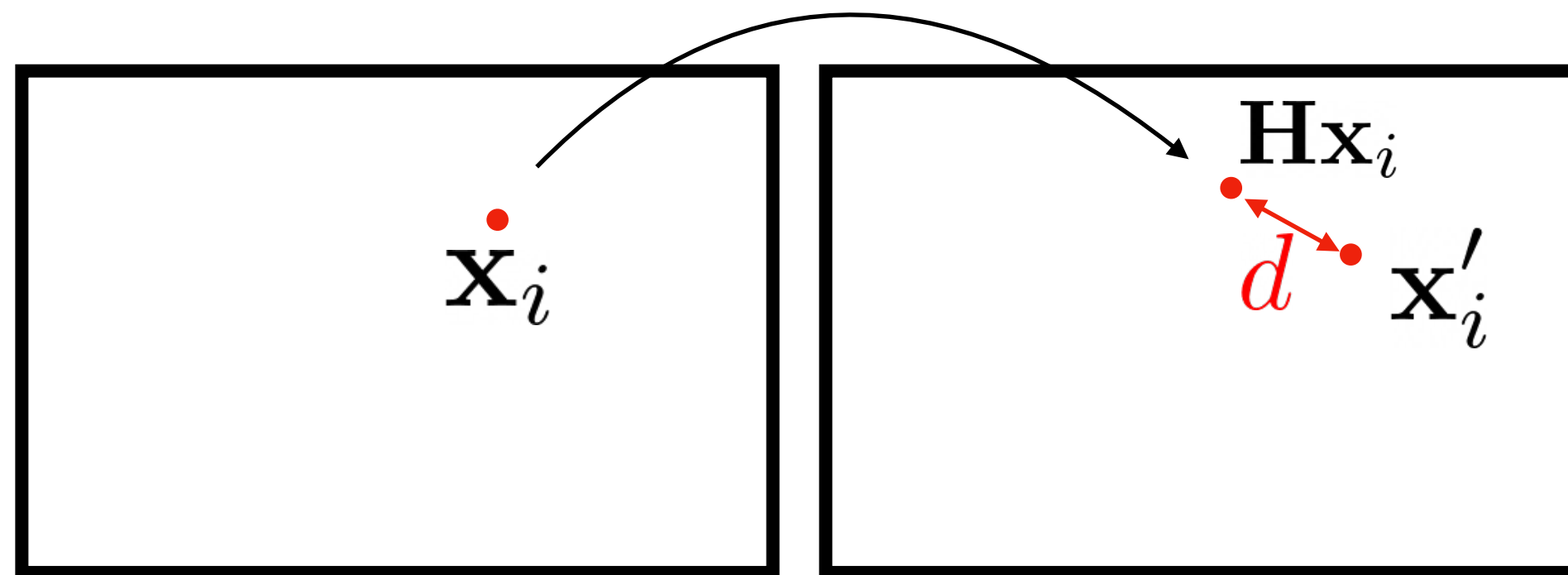
a) Initialize  $\mathbf{H}$  using DLT

b) Iteratively update  $\mathbf{H}$  using the non-linear objective

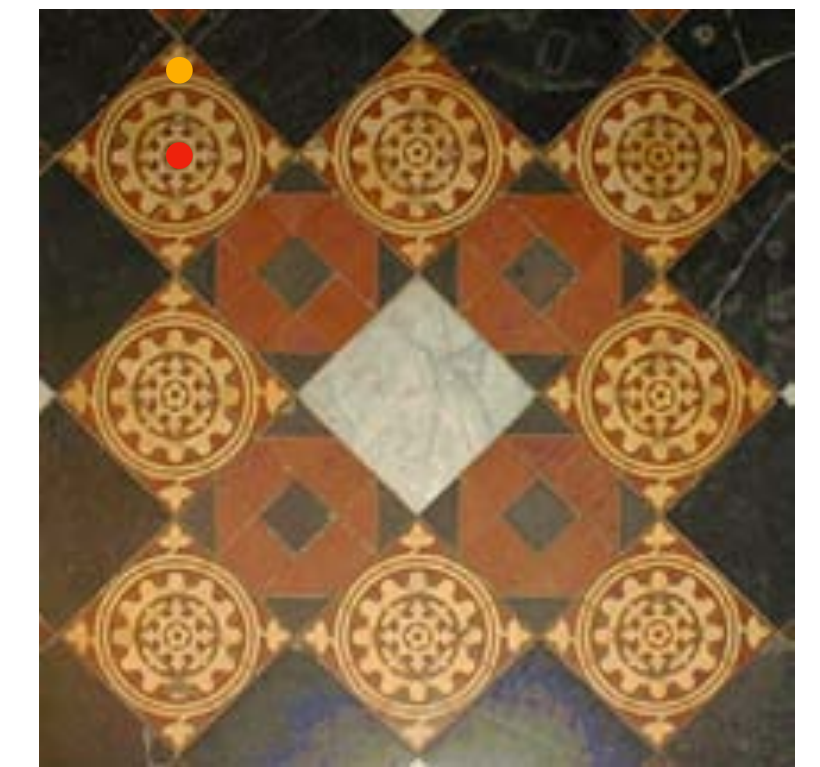
# Beyond DLT

$$L = \sum_i d(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i) \quad \min_H L$$

$$d_{\text{geom}}(\mathbf{x}, \mathbf{x}') = \|x'/w' - x/w\|^2 + \|y'/w' - y/w\|^2$$



Only allows for error in one image!

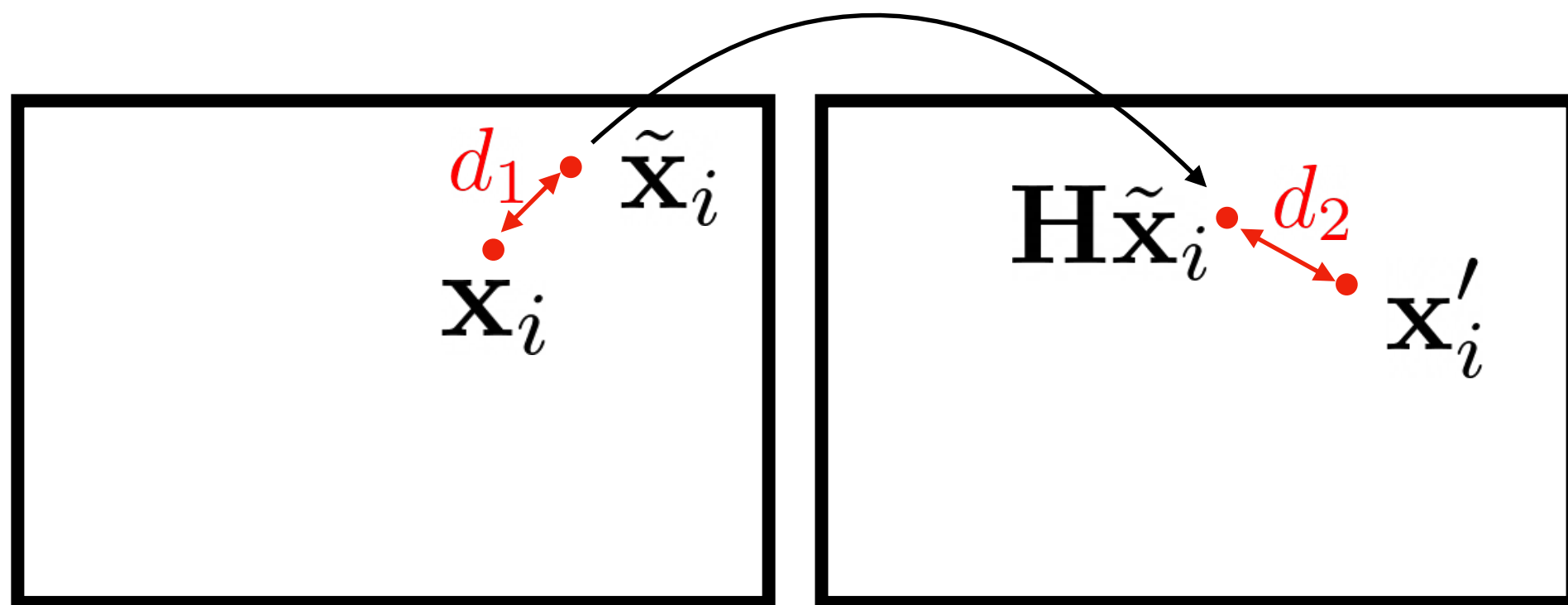


Small annotation error in one image can cause much larger errors in other

# Beyond DLT

$$L = \sum_i d(\tilde{\mathbf{x}}_i, \mathbf{x}_i) + \sum_i d(\mathbf{H}\tilde{\mathbf{x}}_i, \mathbf{x}'_i) \quad \min_{H, \{\tilde{\mathbf{x}}_i\}} L$$

$$d_{\text{geom}}(\mathbf{x}, \mathbf{x}') = \|x'/w' - x/w\|^2 + \|y'/w' - y/w\|^2$$



Need to optimize  $2n + 8$  variables

Q: Do 4 correspondences still suffice?

Allow for error in both images



# Summary

$$L = \sum_i d_{alg}(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i)$$

$$\min_H L$$



$$\min_{\mathbf{h}, \|\mathbf{h}\|=1} \|\mathbf{A}\mathbf{h}\|$$

$$L = \sum_i d_{geom}(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i)$$

$$\min_H L$$



Iterative optimization after initialization

$$L = \sum_i d_g(\tilde{\mathbf{x}}_i, \mathbf{x}_i) + \sum_i d_g(\mathbf{H}\tilde{\mathbf{x}}_i, \mathbf{x}'_i)$$

$$\min_{H, \{\tilde{\mathbf{x}}_i\}} L$$

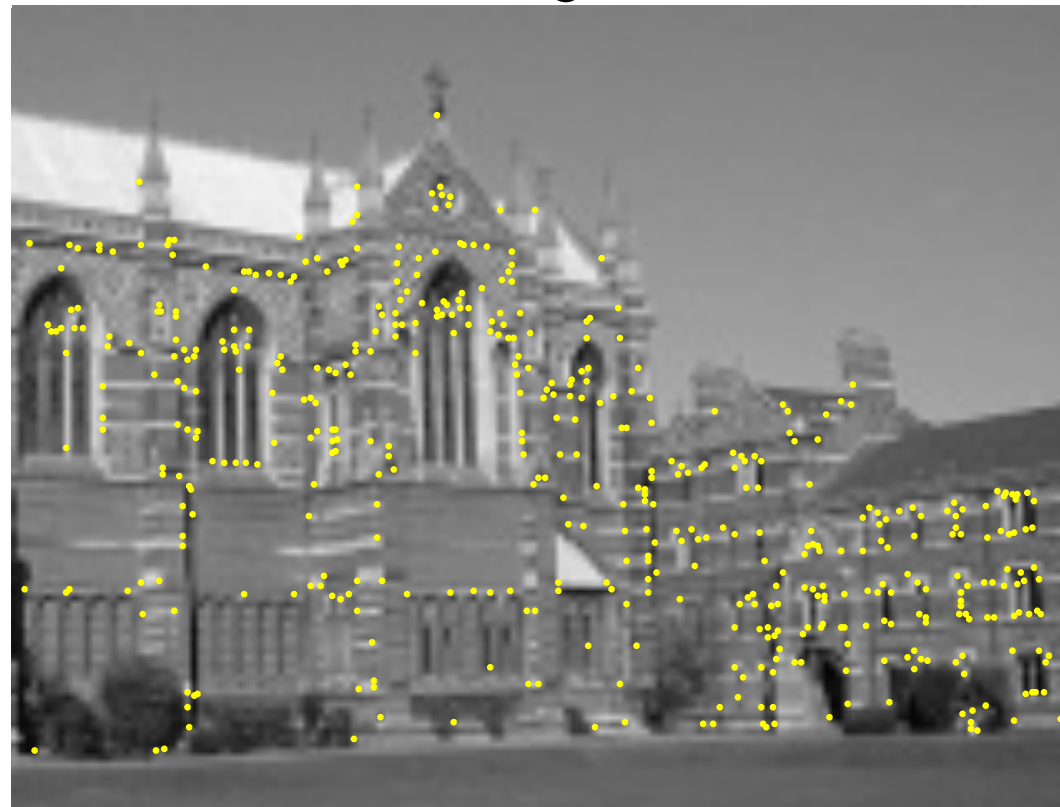
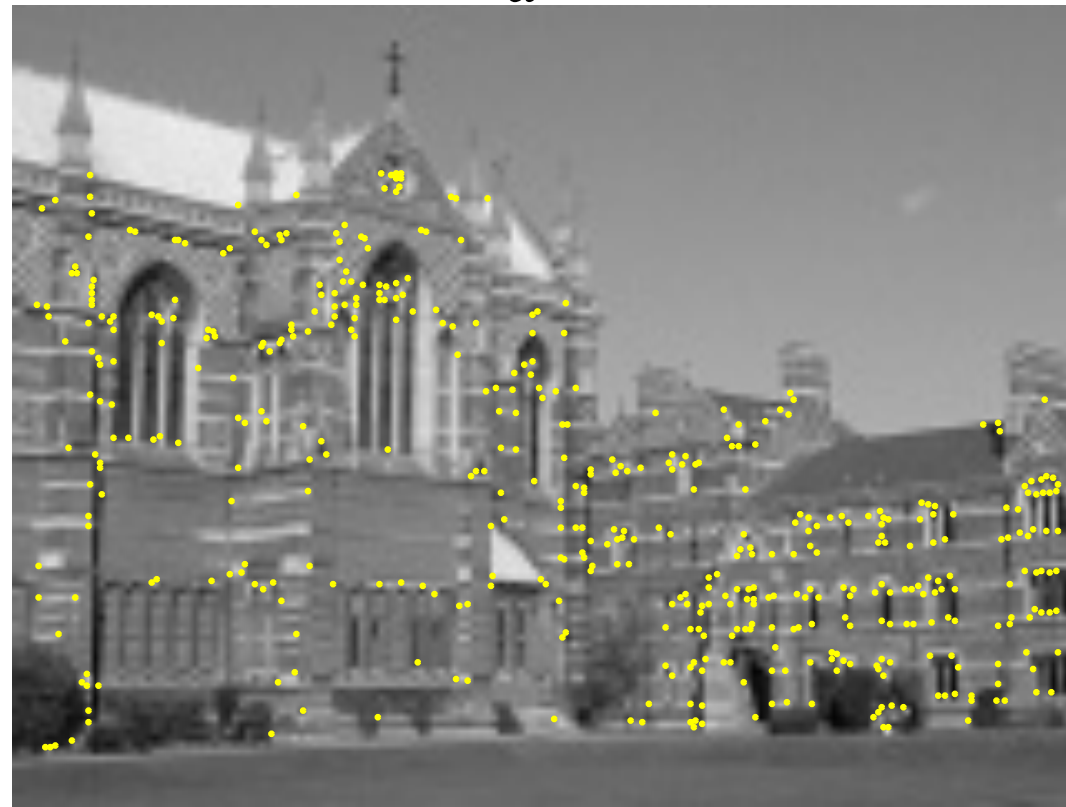


Iterative optimization after initialization

Q: Which objective should I actually use?

Often, the first suffices (but normalization is important!)

# Dealing with Noise

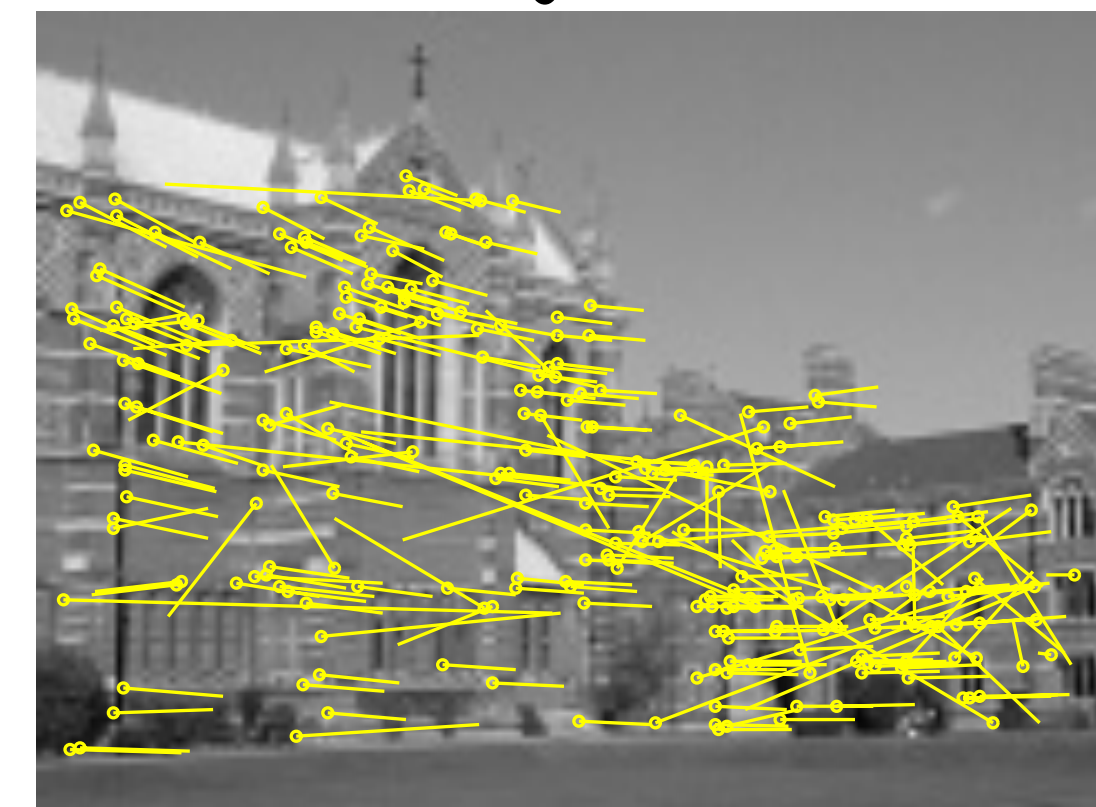
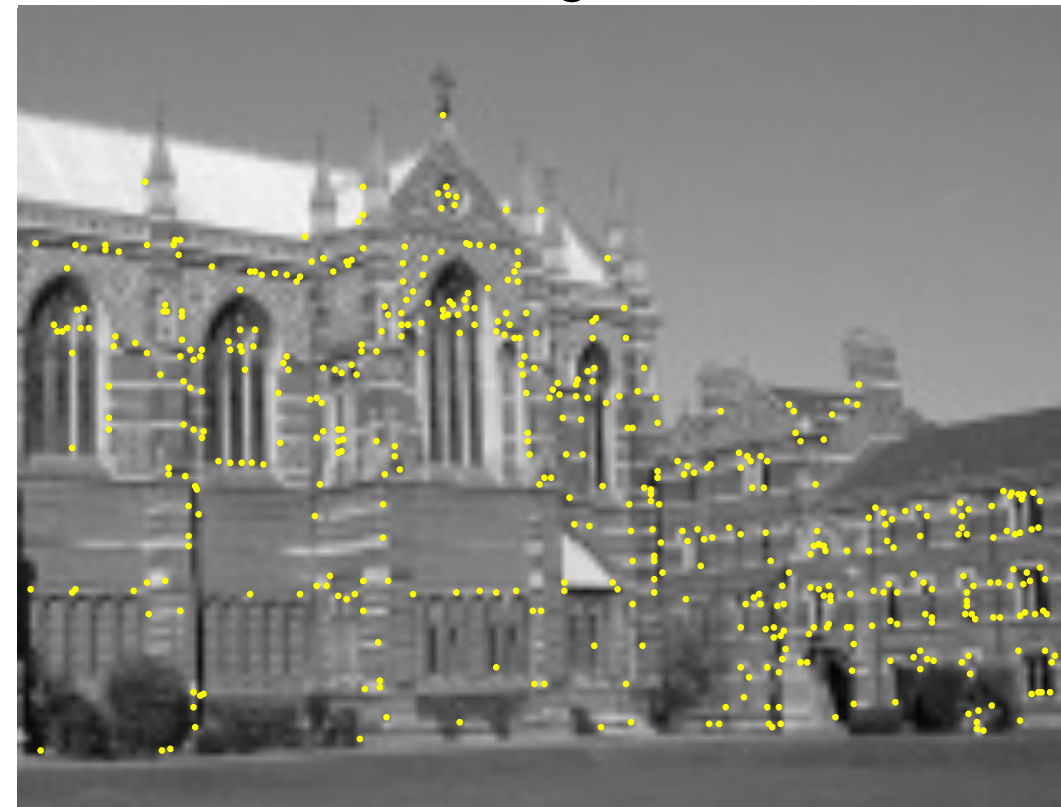
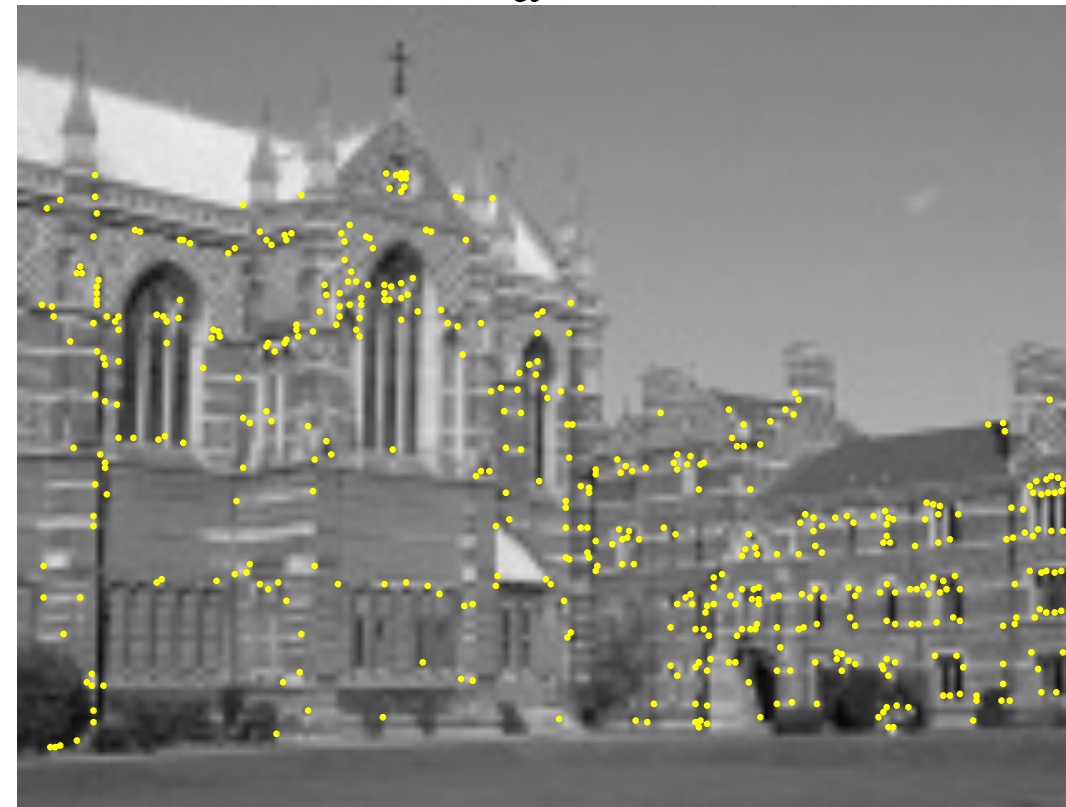


Step 1: Compute interest points and descriptors

Step 2: Find potential matches ( $\mathbf{x}_i, \mathbf{x}_i'$ )

~~Step 3: Find optimal  $\mathbf{x}_i, \mathbf{x}_i'$~~

# Dealing with Noise: RANSAC



Step 1: Compute interest points and descriptors

Step 2: Find potential matches ( $\mathbf{x}_i, \mathbf{x}_i'$ )

Step 3: For N iterations:

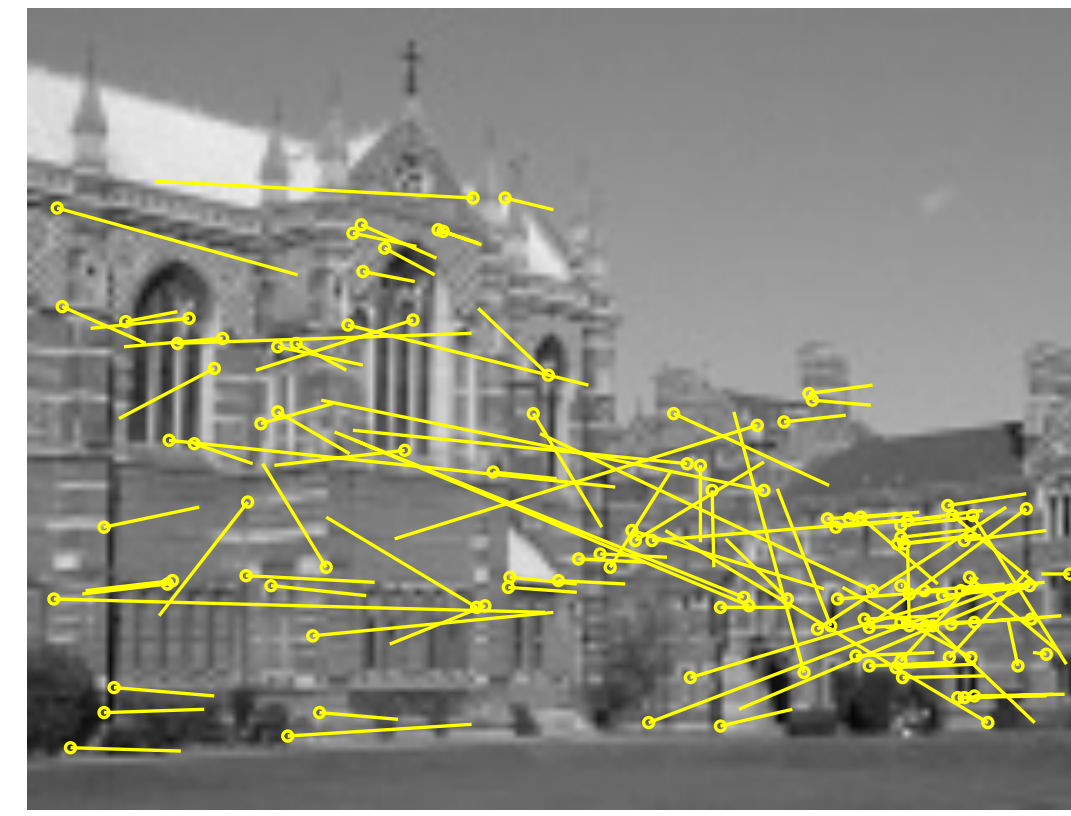
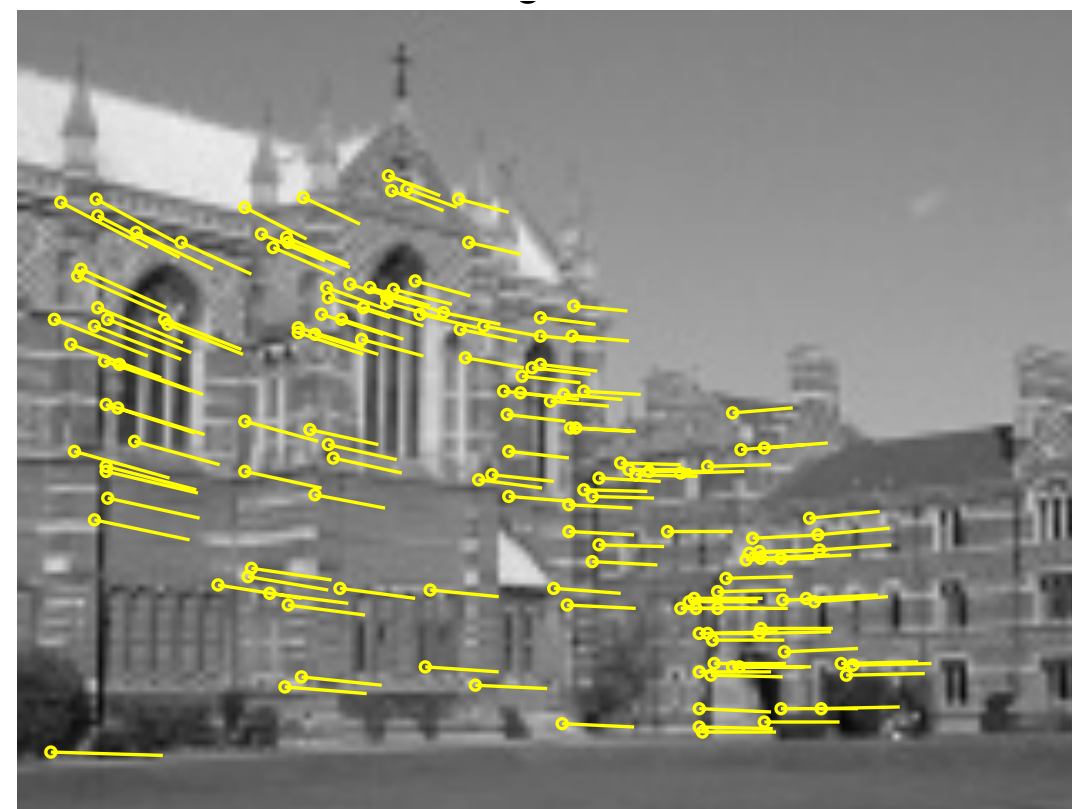
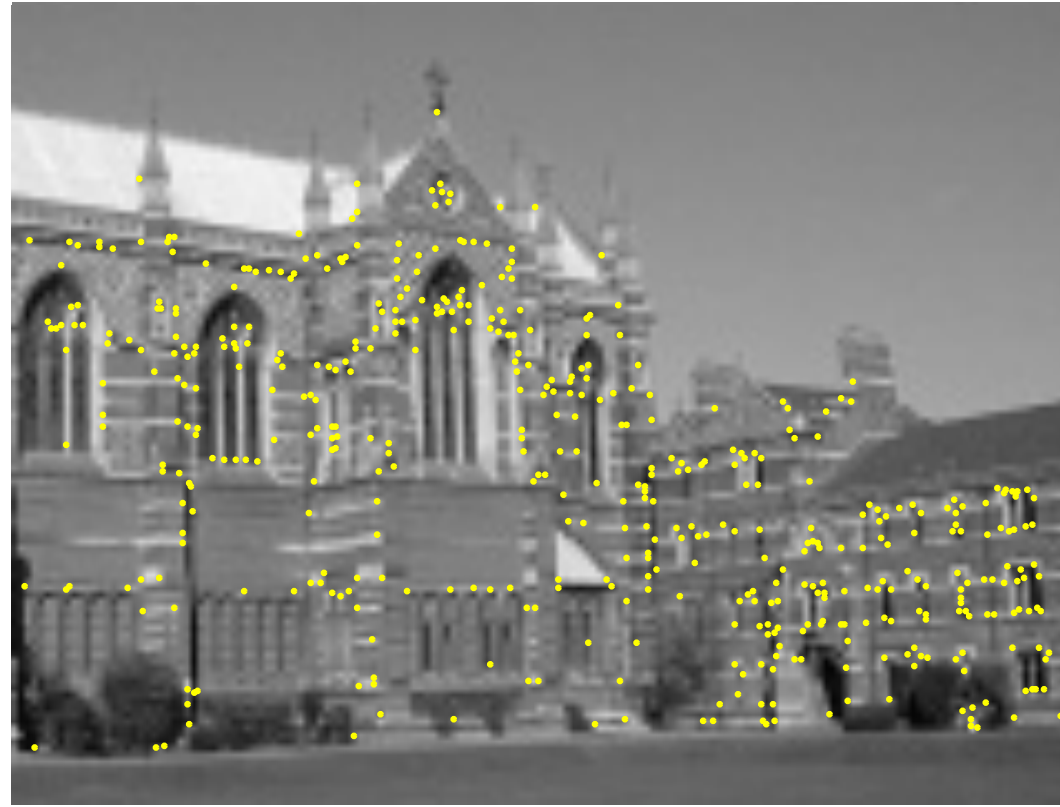
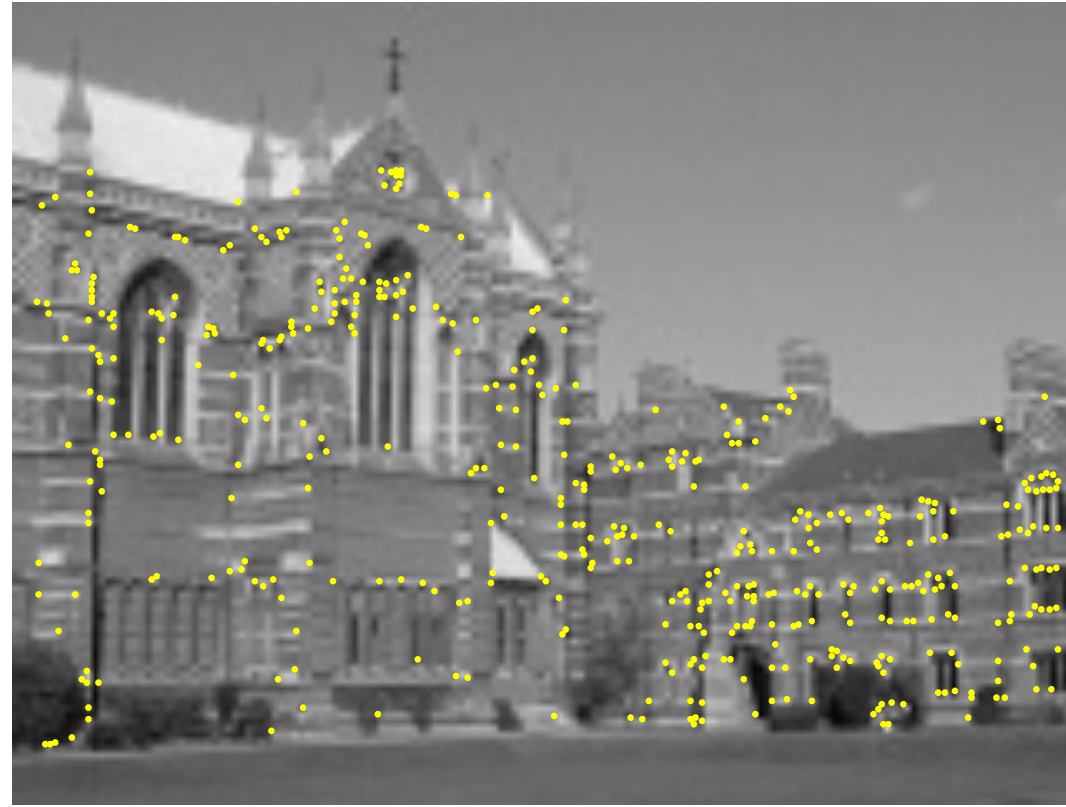
- Sample K potential matches to compute  $\mathbf{H}_n$
- Compute number of inliers (matches with error  $< e$  w.r.t.  $\mathbf{H}_n$ )

Several tunable hyper-parameters

Use the largest set of inliers to compute  $\mathbf{H}$



# Dealing with Noise: RANSAC



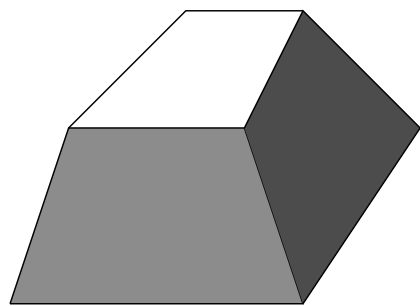
# Projective Transforms from 3D to 2D: Cameras!



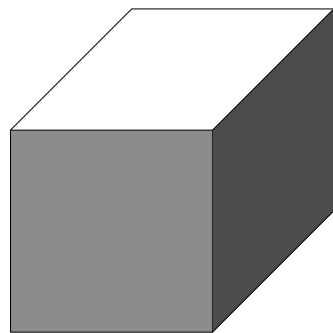
$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



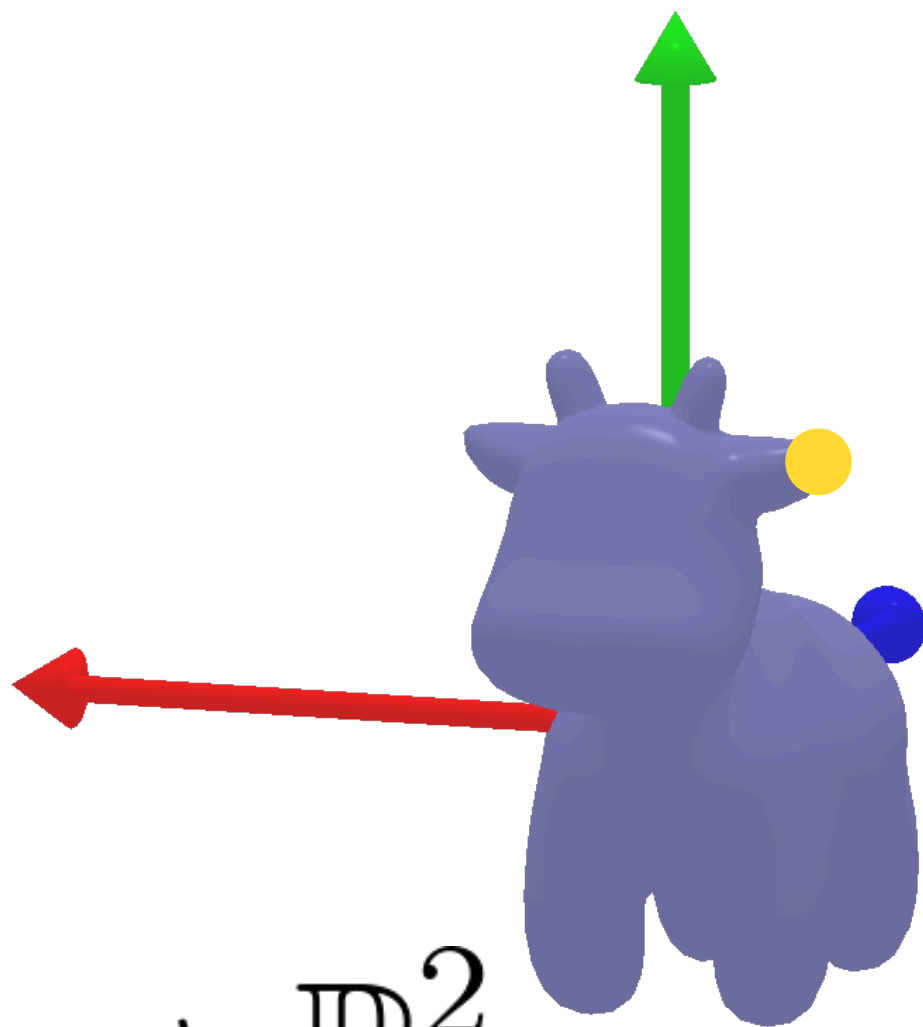
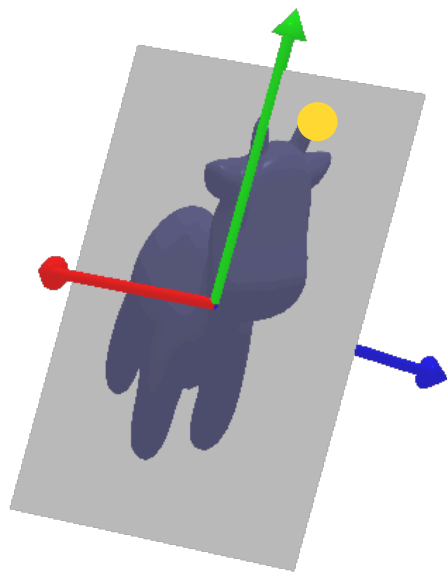
$$\mathbb{P}^2 \rightarrow \mathbb{P}^2$$



$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



$$\mathbb{P}^3 \rightarrow \mathbb{P}^3$$



$$\mathbb{P}^3 \rightarrow \mathbb{P}^2$$

# Questions?